Efficient Vision Transformer Models

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Abstract

This study investigates efficient attention mechanisms within Vision Transformer architectures. A regular Vision Transformer baseline is compared with a Performer-based variant and two structured positional variants, namely Circulant-STRING and Cayley-STRING. Experiments are conducted on CIFAR-10, CIFAR-100, MNIST, and Fashion-MNIST, with evaluation based on validation accuracy, training time, inference time, and accuracy–efficiency trade-offs. The results reveal that efficient attention introduces measurable approximation error, particularly in complex datasets, while structured positional encoding can partially mitigate this degradation. No single model dominates across all datasets, highlighting the importance of task-dependent model selection.

1 Introduction

Vision Transformers model images as sequences of patches and apply self-attention to capture global relationships. This idea builds on the Transformer architecture, which replaces recurrence with attention-based sequence modelling (Vaswani et al., 2017). The Vision Transformer adapts this mechanism to image recognition by treating image patches as input tokens (Dosovitskiy et al., 2020).

Although effective, standard self-attention scales quadratically with the number of tokens. This can become computationally expensive as image resolution or patch sequence length increases (Choromanski et al., 2020). Efficient attention mechanisms have therefore been proposed to reduce computational cost while preserving useful attention behaviour. Among them, Performer replaces softmax attention with a random-feature approximation, while STRING-style positional mechanisms aim to improve spatial representation through structured positional encoding (Choromanski et al., 2020; Schenck et al., 2025).

This study evaluates the following hypothesis:

> Efficient attention mechanisms reduce computational cost but introduce approximation error, while structured positional encoding can improve spatial modelling and partially compensate for this loss.

The objective is to assess whether these expected behaviours hold across datasets of varying complexity. The four model variants compared in this project are a regular ViT baseline, a Performer-based ViT, Circulant-STRING ViT, and Cayley-STRING ViT.

2 Background

2.1 Vision Transformer

A Vision Transformer divides an input image into fixed-size patches, embeds each patch into a vector representation, adds positional information, and processes the resulting token sequence using Transformer encoder blocks (Dosovitskiy et al., 2020). A classification token aggregates global information and is passed into a classification head for final prediction. This architecture follows the broader Transformer principle of using self-attention to model relationships between tokens (Vaswani et al., 2017). Additional tutorial-level explanations of self-attention and encoder–decoder Transformers are provided by Kamper (2024) and von Platen (2020).

2.2 Standard Attention

Standard scaled dot-product attention is the core operation used in the original Transformer architecture (Vaswani et al., 2017). Given query, key, and value matrices Q, K, and V, attention is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

where d denotes the key dimension. The term QK^T computes pairwise similarity scores between all tokens, and the softmax operation converts these scores into attention weights. Because every token is compared with every other token, the attention matrix grows quadratically with sequence length. This quadratic cost is one of the main reasons why efficient attention variants are studied (Choromanski et al., 2020).

2.3 Performer

Performer is an efficient Transformer variant designed to approximate standard softmax attention without explicitly forming the full quadratic attention matrix (Choromanski et al., 2020). Its main mechanism, FAVOR+, uses positive orthogonal random features to approximate the softmax kernel. In simplified form, the softmax kernel between a query vector q and a key vector k can be written as:

$$\exp(q^T k) \approx \phi(q)^T \phi(k)$$

where $\phi(\cdot)$ denotes a random feature mapping. This approximation allows attention to be computed through feature products rather than through the full pairwise attention matrix. As a result, Performer can reduce the computational burden of attention, although the approximation may introduce error when fine-grained attention patterns are important. This trade-off between efficiency and approximation quality is central to the experiments in this project.

2.4 STRING-style Positional Variants

The Circulant-STRING and Cayley-STRING variants are included to test whether structured positional mechanisms can improve spatial modelling in image classification. The original STRING work studies improved two-dimensional and three-dimensional positional encodings for attention models (Schenck et al., 2025). In this project, the two STRING variants should be interpreted as simplified project-level implementations inspired by STRING-style mechanisms, rather than exact reproductions of the original method.

3 Methodology

Four model variants are evaluated under a unified Vision Transformer framework. To ensure a controlled comparison, all models share the same patch embedding module, Transformer encoder depth, hidden dimension, number of attention heads, classification head, and training schedule. They differ only in the attention mechanism or positional encoding component.

Baseline ViT The baseline model uses the standard scaled dot-product attention mechanism from the original Transformer architecture (Vaswani et al., 2017):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

where Q, K, and V represent the query, key, and value matrices, and d denotes the key dimension. This model serves as the reference point for both predictive accuracy and computational cost. It provides the standard ViT-style baseline against which the efficient and position-aware variants are compared.

Performer ViT The Performer ViT replaces explicit softmax attention with a random-feature approximation inspired by the FAVOR+ mechanism (Choromanski et al., 2020). Instead of directly forming the full attention matrix, the softmax kernel is approximated by feature mappings:

$$\exp(q^T k) \approx \phi(q)^T \phi(k)$$

where $\phi(\cdot)$ denotes a random feature map. This allows attention to be computed more efficiently without explicitly storing the full quadratic attention matrix. In this project, the Performer model is used to test whether an efficient attention approximation can reduce computational cost while maintaining competitive classification accuracy.

Circulant-STRING ViT The Circulant-STRING ViT incorporates structured relative positional information inspired by the circulant-STRING mechanism (Schenck et al., 2025). The attention logits are modified by adding a positional bias term:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V$$

where B is a learnable positional bias indexed by cyclic relative offsets:

$$\Delta(i, j) = (i - j) \mod N$$

This design allows the model to encode relative spatial relationships between patch tokens while keeping the overall ViT structure unchanged. The purpose of this variant is to examine whether structured positional information improves image classification performance compared with the standard ViT and Performer variants.

Cayley-STRING ViT The Cayley-STRING ViT uses a structured positional transformation inspired by the Cayley-STRING mechanism (Schenck et al., 2025). The transformation is represented as:

$$R = (I - A)(I + A)^{-1}$$

where A is a skew-symmetric matrix. This construction produces a structured transformation that can be used to represent relative positional information in the attention computation.

Compared with the circulant version, the Cayley-style variant introduces a richer positional transformation, but it can also increase computational overhead.

Implementation Note The model variants follow the structure required in the project brief: regular ViT, Performer ViT, Circulant-STRING ViT, and Cayley-STRING ViT. The baseline ViT and Performer ViT follow the standard attention and Performer-style efficient attention formulations, respectively. The two STRING variants implement structured relative positional mechanisms inspired by Schenck et al. (Schenck et al., 2025) within the same ViT pipeline. This keeps the experimental comparison controlled, since the observed differences mainly come from the attention or positional components rather than from unrelated architectural changes.

4 Experimental Setup

4.1 Datasets

The experiments use four image classification datasets:

- CIFAR-10
- CIFAR-100
- MNIST
- Fashion-MNIST

These datasets span multiple levels of complexity. CIFAR-10 provides moderate complexity with natural images. CIFAR-100 introduces higher difficulty due to increased class granularity and inter-class similarity. MNIST represents a low-complexity setting with simple spatial structure, while Fashion-MNIST introduces moderate variation in texture and shape.

4.2 Hyperparameters

All models use identical settings:

- Image size: 32×32
- Patch size: 4
- Embedding dimension: 128
- Depth: 4
- Heads: 4
- Epochs: 5

This configuration ensures a controlled comparison under a limited computational budget. Short training schedules are used to highlight relative differences between models rather than to maximise absolute performance.

4.3 Evaluation Metrics

Evaluation includes:

- Validation accuracy
- Training time
- Inference time

In addition, we visualise the trade-off between validation accuracy and training time. This is useful because a model with slightly higher accuracy may not be practically preferable if it requires substantially longer training time.

5 Results

Table 1 summarises the validation accuracy, training time, and inference time for all model variants across the four datasets. Values are reported as mean ± standard deviation over three independent runs.

Table 1: Validation accuracy, training time, and inference time (mean ± std, 3 runs). Best values within each dataset are shown in bold.

Dataset	Model	Validation Accuracy	Training Time (s)	Inference Time (s)
CIFAR-10	Baseline	0.477 ± 0.010	**38.85 ± 0.18**	**5.26 ± 0.02**
	Performer	0.457 ± 0.013	46.03 ± 0.13	5.37 ± 0.07
	Circulant-STRING	**0.478 ± 0.005**	41.08 ± 0.08	5.29 ± 0.06
	Cayley-STRING	0.476 ± 0.014	57.15 ± 0.56	5.44 ± 0.07
CIFAR-100	Baseline	0.158 ± 0.001	**38.67 ± 0.12**	5.39 ± 0.20
	Performer	0.139 ± 0.009	47.54 ± 0.41	5.49 ± 0.09
	Circulant-STRING	**0.166 ± 0.007**	42.00 ± 0.58	**5.33 ± 0.02**
	Cayley-STRING	0.159 ± 0.003	57.89 ± 0.71	5.43 ± 0.07
MNIST	Baseline	0.901 ± 0.010	**35.62 ± 0.19**	**3.05 ± 0.02**
	Performer	**0.916 ± 0.007**	42.62 ± 0.25	3.16 ± 0.05
	Circulant-STRING	0.904 ± 0.017	38.05 ± 0.11	3.14 ± 0.16
	Cayley-STRING	0.906 ± 0.009	53.74 ± 0.42	3.24 ± 0.04
Fashion-MNIST	Baseline	0.789 ± 0.014	**35.70 ± 0.10**	**3.09 ± 0.03**
	Performer	**0.797 ± 0.005**	42.82 ± 0.08	3.15 ± 0.07
	Circulant-STRING	0.795 ± 0.005	38.60 ± 0.84	3.12 ± 0.15
	Cayley-STRING	0.791 ± 0.011	54.24 ± 0.27	3.22 ± 0.01

5.1 Validation Accuracy

Figure 1 reports the validation accuracy comparison across models and datasets. The figure includes mean and standard deviation values, allowing the comparison to account for variability across runs where applicable.

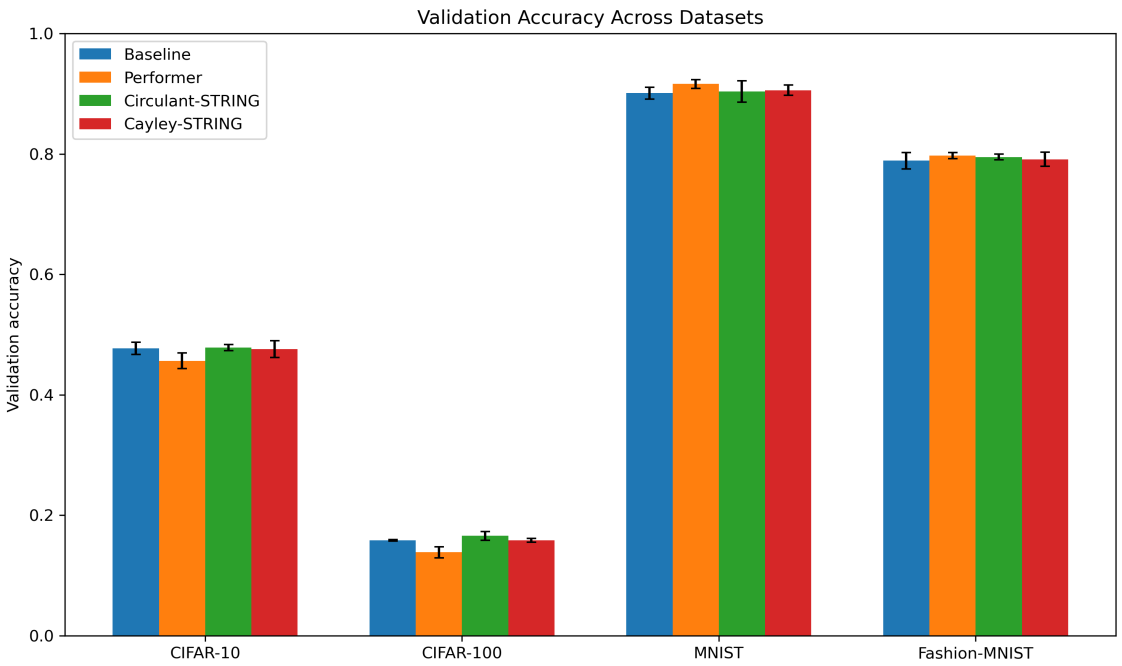


Figure 1: Validation accuracy comparison across datasets and model variants.

The results indicate clear dataset-dependent behaviour. On CIFAR-10, the Performer exhibits a consistent accuracy drop of approximately 3–5% relative to the baseline, indicating that approximation error affects moderately complex tasks. On CIFAR-100, the performance gap increases further, suggesting that approximation error becomes more damaging in complex classification settings. In contrast, the Circulant-STRING variant achieves the highest accuracy in several settings, suggesting that structured positional encoding can improve spatial modelling. On MNIST and Fashion-MNIST, Performer achieves comparable or higher accuracy, indicating that approximation error is less significant in low-complexity tasks.

5.2 Training Time

Figure 2 shows the training time comparison across datasets and model variants.

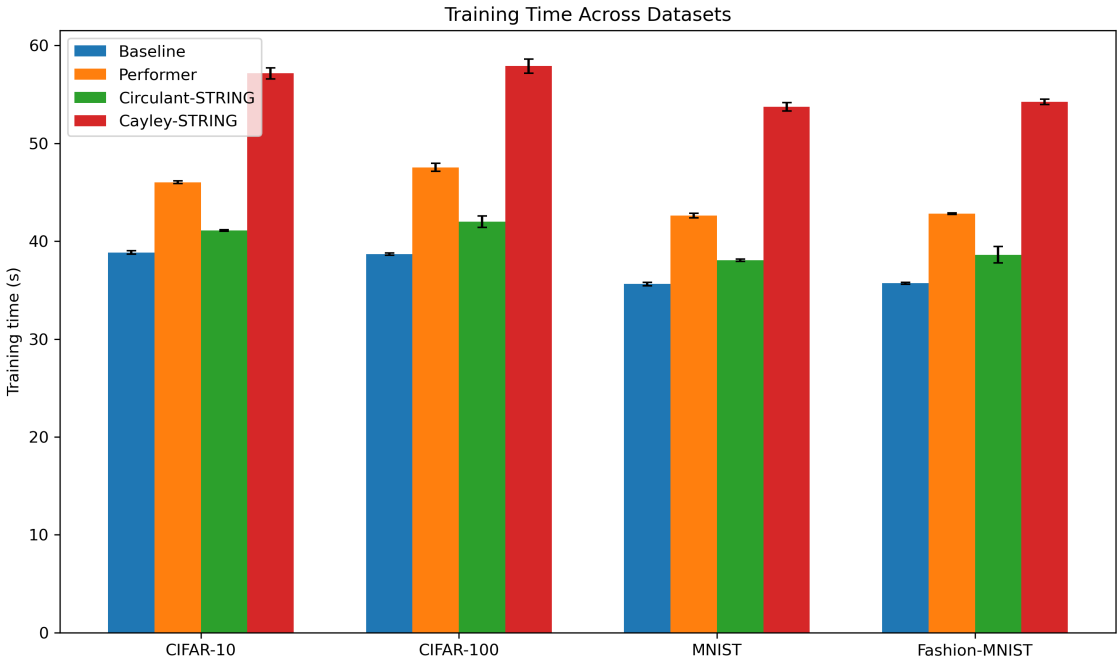


Figure 2: Training time comparison across datasets and model variants.

Although Performer reduces the theoretical complexity of attention, the empirical training time results reveal a different pattern. Across all datasets, the baseline ViT consistently achieves the lowest training time, while the Performer-based model is slower by approximately 15–20%.

For example, on CIFAR-10 and CIFAR-100, the baseline model requires around 38–39 seconds, whereas Performer increases the training time to approximately 46–48 seconds. Similar trends are observed on MNIST and Fashion-MNIST, where Performer remains consistently slower than the baseline.

The structured positional variants further increase computational cost. Circulant-STRING introduces a moderate overhead of approximately 5–10% compared to the baseline, while Cayley-STRING exhibits the highest training time across all datasets, reaching roughly 54–58 seconds. This suggests that the additional matrix transformations and positional computations introduce non-negligible computational overhead.

These results indicate that theoretical improvements in asymptotic complexity do not necessarily translate into practical speedups in small-scale settings. The overhead associated with random feature computation in Performer, as well as additional positional operations in STRING variants, dominates the runtime when sequence length is relatively small.

Overall, the results demonstrate that implementation overhead and constant factors play a critical role in determining real-world performance, particularly when model and dataset sizes are limited.

5.3 Inference Time

Figure 3 reports inference time across the four datasets.

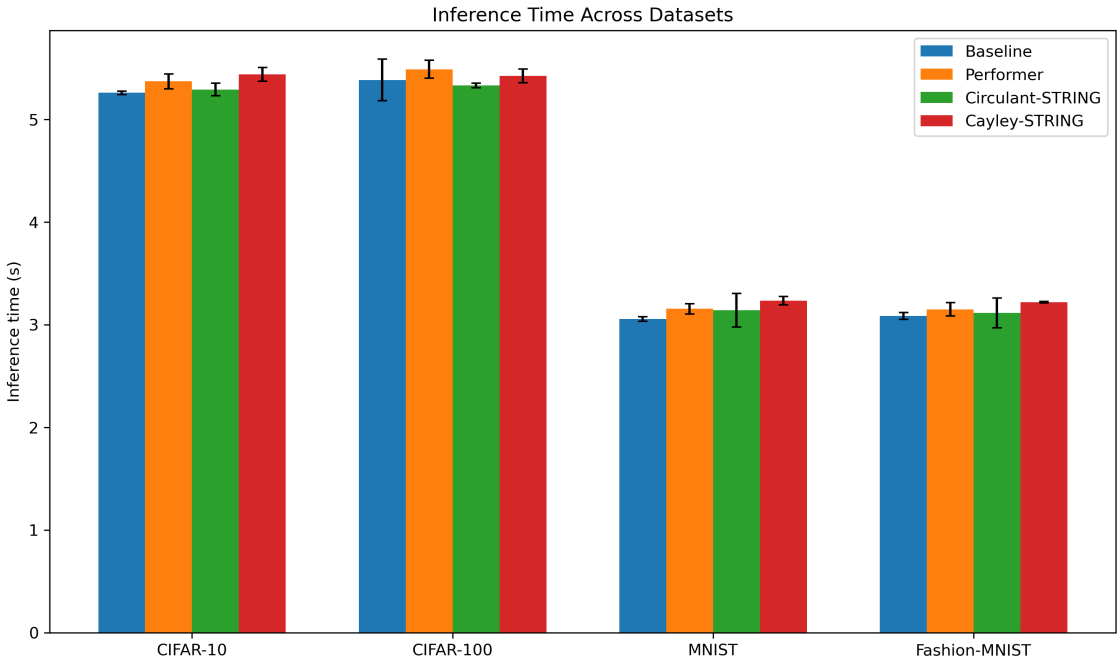


Figure 3: Inference time comparison across datasets and model variants.

The inference-time comparison reveals that efficient attention mechanisms do not consistently yield faster inference in practice. Across all datasets, the Performer-based model exhibits similar or slightly higher inference latency compared to the baseline, with increases on the order of 5–15%.

This indicates that the computational overhead introduced by random feature transformations and additional kernel operations offsets the theoretical reduction in complexity. Similarly, the STRING-based variants further increase inference time due to added positional computations, with Cayley-STRING consistently incurring the highest latency.

These results suggest that, in small-scale image classification settings where sequence lengths are limited, constant factors and implementation overhead dominate asymptotic complexity. As a result, the theoretical efficiency advantages of linear attention mechanisms are not fully realized in practical inference scenarios.

5.4 Accuracy–Training Time Trade-off

Figure 4 summarises the overall relationship between validation accuracy and training time across all datasets.

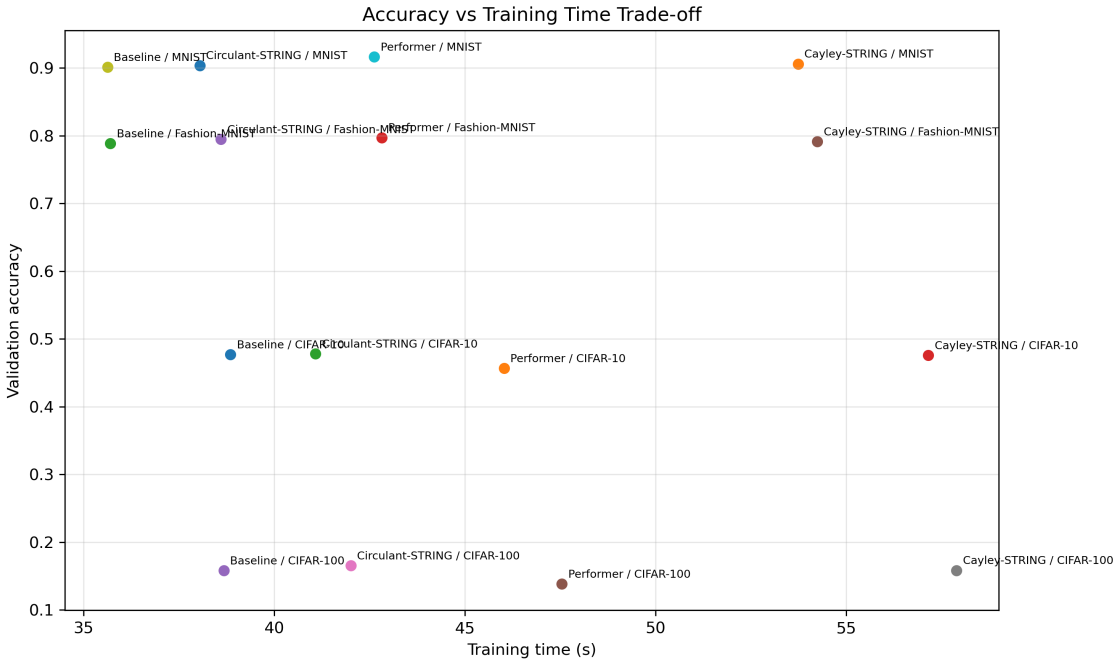


Figure 4: Overall trade-off between validation accuracy and training time.

The trade-off plot gives a more balanced view than accuracy alone. A model located higher and further left is preferable because it achieves stronger accuracy with lower training cost. The results show a clear separation between dataset difficulty levels. MNIST and Fashion-MNIST points are concentrated in the upper part of the figure, indicating that all models achieve relatively high validation accuracy on simpler greyscale tasks. In contrast, CIFAR-10 and especially CIFAR-100 appear much lower on the accuracy axis, confirming that the colour-image datasets are more challenging under the current short training schedule.

The figure also shows that theoretical efficiency does not directly translate into empirical speed advantage in this setting. Although Performer is designed to reduce the quadratic cost of attention, its points are not consistently located further left than the baseline. This suggests that, for small image sizes and short token sequences, the overhead of random-feature attention can offset its theoretical benefit. Similarly, Cayley-STRING often requires noticeably longer training time without producing a proportionate improvement in accuracy, particularly on CIFAR-10 and CIFAR-100.

Among the structured positional variants, Circulant-STRING appears to offer a more favourable compromise. It tends to remain closer to the baseline in training time while sometimes improving or maintaining validation accuracy. This makes it more attractive than Cayley-STRING in this small-scale experimental setting. Overall, the main observation is that no single model dominates all datasets. Efficient and structured attention methods should therefore be evaluated under task-specific conditions rather than assumed to be universally superior.

5.5 Dataset-specific Trade-off Analysis

Figures 5–8 show the accuracy–training time trade-off separately for each dataset.

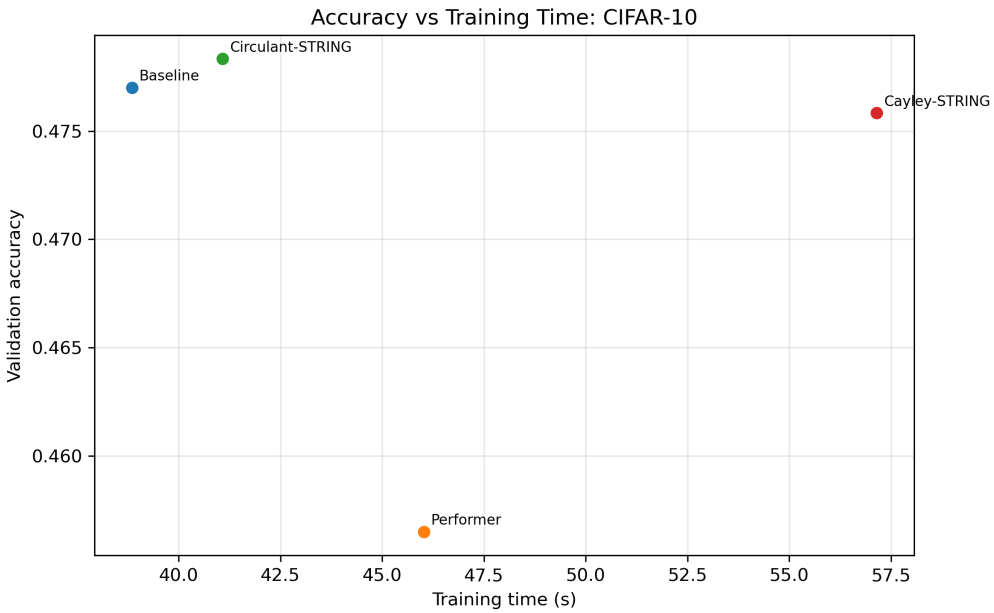


Figure 5: Accuracy–training time trade-off on CIFAR-10.

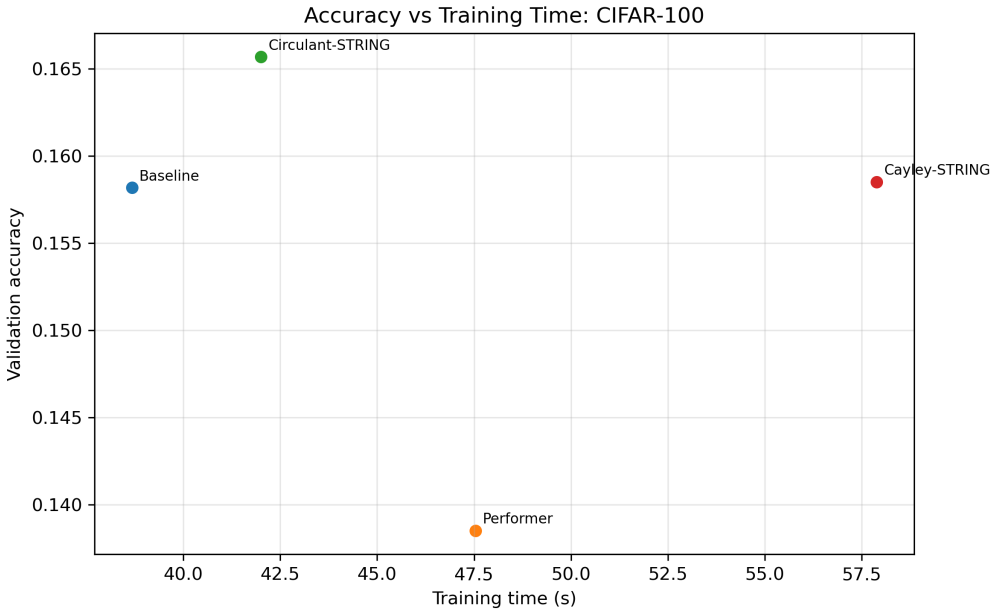


Figure 6: Accuracy–training time trade-off on CIFAR-100.

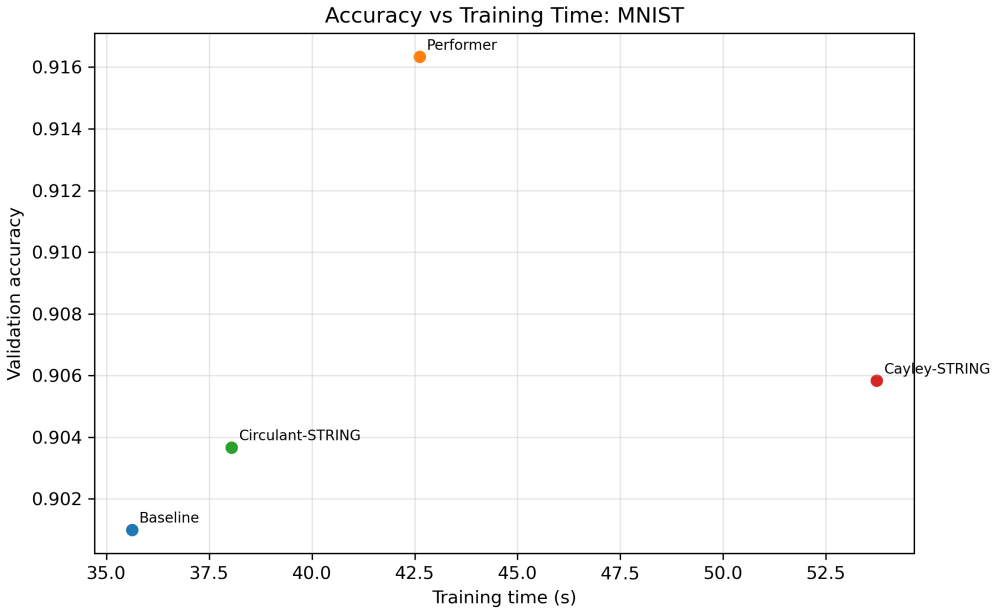


Figure 7: Accuracy–training time trade-off on MNIST.

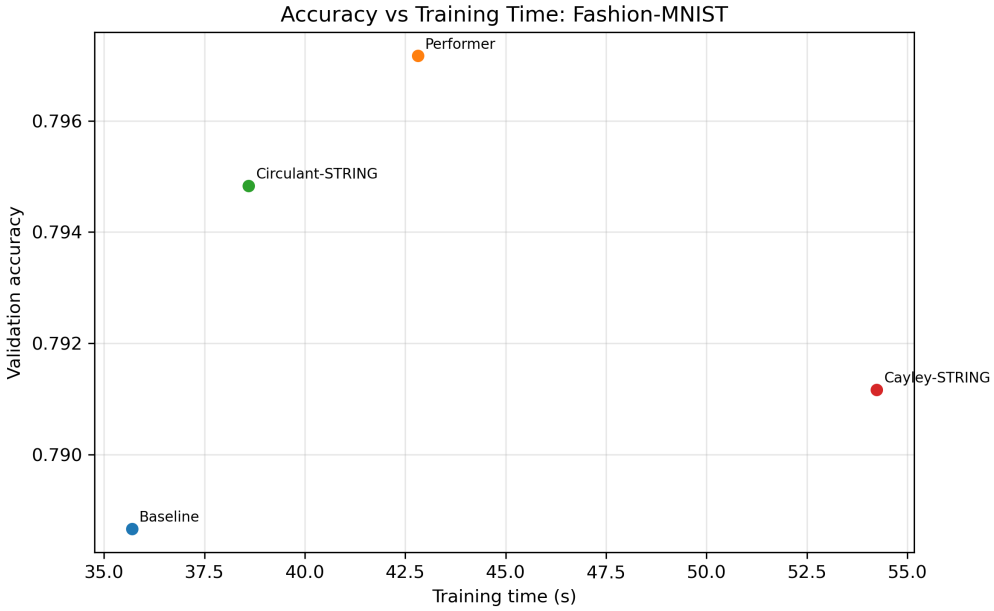


Figure 8: Accuracy–training time trade-off on Fashion-MNIST.

Across the dataset-specific trade-off plots, the key pattern is that model behaviour changes substantially across datasets. On CIFAR-10, the baseline and Circulant-STRING models are close in both accuracy and training time, while Circulant-STRING gives a slight accuracy improvement with only a small increase in cost. In contrast, Performer records lower accuracy, and Cayley-STRING requires much longer training time without a clear accuracy advantage. This suggests that, for CIFAR-10, the Circulant-STRING variant provides the most favourable balance between accuracy and efficiency.

On CIFAR-100, all models achieve much lower validation accuracy than on CIFAR-10, reflecting the greater difficulty of the dataset. Circulant-STRING again performs best in accuracy, while the baseline remains the fastest and still competitive. Performer performs weakest,

suggesting that approximation error is more harmful when the classification task is more fine-grained. Cayley-STRING does not justify its additional training time because its accuracy gain over the baseline is very small.

The MNIST and Fashion-MNIST plots show a different pattern. On these simpler greyscale datasets, Performer achieves the highest validation accuracy, indicating that the random-feature approximation is less damaging when the visual structure is simpler. However, Performer is not the fastest model, so its advantage is mainly predictive rather than computational. The baseline remains the most efficient option, but it also gives the lowest accuracy. Circulant-STRING offers a middle-ground solution, improving over the baseline while keeping training time moderate.

Overall, the dataset-specific plots confirm that no model is universally optimal. Performer is more suitable for simpler datasets, Circulant-STRING gives the most stable trade-off across datasets, and Cayley-STRING appears relatively expensive under the current experimental setting. This supports the broader conclusion that efficient attention methods should be evaluated empirically rather than selected only from theoretical complexity arguments.

6 Discussion

The experimental results show that the behaviour of efficient attention mechanisms is strongly dependent on dataset complexity and implementation context. Although Performer is theoretically designed to reduce the asymptotic cost of attention by avoiding explicit construction of the full quadratic attention matrix, this advantage is not realised in the present small-scale image classification setting. The patch sequence length is relatively short, and the additional overhead from random feature computation offsets the theoretical efficiency gain. As a result, the baseline ViT remains the fastest model in both training and inference across most datasets.

The accuracy results also suggest that approximation error has different effects across datasets. On CIFAR-10 and CIFAR-100, Performer generally performs worse than the baseline, with the gap becoming more visible on CIFAR-100. This is expected because CIFAR-100 is a more fine-grained classification task with more classes and greater inter-class similarity. In such settings, small errors in approximating attention weights can accumulate across Transformer layers and reduce the model's ability to capture useful relationships between image patches. Therefore, Performer is not uniformly better than standard attention in this experiment, even though it has a stronger theoretical efficiency profile.

The results on MNIST and Fashion-MNIST show a different pattern. On these simpler greyscale datasets, Performer achieves competitive or higher validation accuracy. This suggests that random-feature attention approximation is less damaging when the visual structure is simpler and the classification task depends on clearer spatial patterns. However, this improvement should not be interpreted as an efficiency advantage, because Performer still requires longer training time than the baseline under the current implementation.

Among the two STRING-inspired variants, Circulant-STRING provides the most balanced performance. It often improves or maintains validation accuracy while introducing only moderate additional training cost. This indicates that structured relative positional information can help the model capture spatial relationships between image patches. In contrast, Cayley-STRING usually introduces the largest computational overhead and does not consistently provide enough accuracy improvement to justify this cost. In this project setting, the circulant variant is therefore more practical than the Cayley-style variant.

It is important to emphasise that the STRING-based models in this project are simplified implementations inspired by STRING-style positional mechanisms rather than exact reproductions of the full STRING method. They are implemented to test whether structured relative positional bias can improve a ViT-style image classifier under a controlled experimental pipeline. This design choice makes the comparison feasible within the project scope, but it also limits the strength of claims that can be made about the original STRING architecture.

Overall, the findings support a more cautious interpretation of efficient Transformer design. Efficient attention mechanisms should not be selected only because of theoretical complexity. Their actual performance depends on sequence length, implementation overhead, dataset difficulty, and the specific form of positional information used. In this experiment, the baseline ViT remains the most computationally efficient option, Performer is more suitable for simpler datasets where approximation error is less harmful, and Circulant-STRING offers the strongest overall trade-off between accuracy and cost.

Several limitations should be acknowledged. First, the experiments use a relatively small-scale training regime with short training schedules, so the absolute validation accuracies should not be interpreted as fully optimised ViT performance. Second, the short patch sequence length limits the potential advantage of linear attention. Performer may show stronger runtime benefits in larger-resolution settings with longer token sequences. Third, the STRING-inspired variants are simplified project-level implementations, and a full reproduction of the original method would require a more complex architecture and more extensive tuning. These limitations mean that the results are best understood as a controlled small-scale comparison rather than a definitive evaluation of all efficient attention models.

7 Conclusion

This project compared four Vision Transformer variants for image classification: a regular ViT baseline, a Performer-based ViT, a Circulant-STRING-inspired ViT, and a Cayley-STRING-inspired ViT. The models were evaluated on CIFAR-10, CIFAR-100, MNIST, and Fashion-MNIST using validation accuracy, training time, inference time, and accuracy–efficiency trade-off analysis.

The results show that no single model dominates across all datasets. The baseline ViT is the most computationally efficient model in the current experimental setting, achieving the lowest training time and generally low inference time. Although Performer has a theoretical advantage in attention complexity, this advantage is not observed empirically here because the sequence length is small and the random-feature overhead dominates runtime. Performer also suffers accuracy degradation on the more complex CIFAR datasets, especially CIFAR-100, but performs competitively on simpler greyscale datasets.

The structured positional variants show that positional information can improve the accuracy–efficiency trade-off, but the effect depends on the specific mechanism. Circulant-STRING provides the most stable compromise, often improving or maintaining accuracy with only moderate additional cost. Cayley-STRING, however, introduces a larger computational overhead and does not consistently produce proportional accuracy gains.

The main conclusion is that efficient attention is not automatically more efficient in practical small-scale ViT experiments. Theoretical complexity must be evaluated together with actual runtime, implementation overhead, and dataset complexity. For this project, the baseline ViT remains the strongest efficiency benchmark, Performer is useful mainly when approximation error is acceptable, and Circulant-STRING is the most promising structured positional variant. Future work should test these models on larger images, longer patch sequences, longer training schedules, and more faithful Performer-STRING implementations to better assess whether the theoretical benefits emerge at scale.

7.1 Code Pointers and Reproducibility

To support reproducibility, the project code is organised into a clear folder structure. The main experiment script is provided in `code/main_experiments.py`. This file controls dataset loading, model construction, training, evaluation, result logging, result aggregation, and command-line execution. The four model implementations are stored separately in the `code/models/` folder:

`baseline_vit.py`, `performer_vit.py`, `circulant_string_vit.py`, and `cayley_string_vit.py`. The plotting functions used to regenerate the figures are provided in `code/plot_results.py`.

The raw experimental results are saved in `results/results.csv`. The generated visualisations used in the report are saved in `results/figures/`. The project also includes `README.txt`, which provides the running instructions, and `requirements.txt`, which lists the required Python packages.

The full experiment can be reproduced by installing the dependencies and running the main script:

```
pip install -r requirements.txt
cd code
python main_experiments.py
```

If the raw results have already been generated, the tables and figures can be regenerated without retraining the models by using:

```
cd code
python main_experiments.py --aggregate-only
```

A small test run is also provided to check whether the code works correctly before running the full experiment:

```
cd code
python main_experiments.py --datasets mnist --models baseline \
--seeds 0 --epochs 1 --train-size 256 --val-size 128
```

For experimental consistency, the code uses fixed random seeds, deterministic dataset subsets, shared hyperparameters, and the same training and evaluation pipeline across all model variants. This ensures that the comparison mainly reflects differences in the attention or positional components rather than unrelated differences in data sampling or training configuration.

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